

Analysis -The Witness

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1. Project Description

For this challenge, I'm going to develop a 2D Real Top-Down game in Godot. The core idea is: **'The Witness'** (the protagonist and title) was in the wrong place at the wrong time and saw something they shouldn't have (a murder, a dirty deal, etc.) in a 1960s jazz club. Now, you have to escape the building before the gangsters catch you, and above all, evade the Boss's right-hand man (**The Hunter**).

This challenge combines two dynamics: **stealth** (guards and gangsters patrolling and detecting you through their line of sight) and **pursuit** (a special NPC, **The Hunter**, who learns to intercept the player increasingly efficiently). For the visual direction, It will be generated by AI inspired from Frank-Miller-inspired noir comic-book aesthetic, utilizing a single 3-color palette (black and white + red accents). This greatly simplifies the asset design and saves time, allowing me to fully focus on the creation process within Godot and Python.

- What I want to learn through this project:

- **Game Development Fundamentals:** I want to learn how to create a video game using an actual industry engine, mastering the core fundamentals, the underlying programming, scene structures, character handling, and collision matrices.
- **Conceptual & Aesthetic Design:** Understand the rationale behind game design choices, specifically, why to choose a particular art style over another, how to build a cohesive narrative, and how to articulate a complete game concept.
- **AI Applied to Video Games via Reinforcement Learning (Q-learning):** Explore how an autonomous agent can learn to behave and improve its performance (specifically for 'The Hunter' role) through trial and error, relying on true algorithmic adaptation rather than a pre-programmed, static finite state machine.
- **Academic Documentation & Empirical Validation:** Learn how to document and validate the entire learning process in a realistic and measurable way, establishing clear metrics to provide concrete proof of whether the agent's efficiency actually improved or not.

2. Requirements (MoScoW)

 Must Have	 Should Have	 Could Have	 Will Not Have
Basic Movement 2D (WASD)	Stamina / Sprint System	Hiding Spots / Hiding Mechanics	Advanced settings menu (Audio, adjustments, graphic settings)
Collisions with the environment	Detection Meter / Alert Gauge	Direct Combat / Killing The Enemy	Multiple levels
Basic Vision Cone for enemies	Search State	Distraction Items	Multiplayer
Playable Prototype	Simple User Interface (Main, Menu and Game Over screens)	Adaptive Sounds Effects	Multiple maps or vertical floors
Fail On Capture / Death State		Q-Learning Prototype (Reinforcement Learning integrated for The Hunter/NPCs)	Crafting
Small Map with Obstacles			Adaptive Jazz Music
NPCs (Enemies)			
NavigationAgent2D Pathfinding			

3. Proposed Solution

- Engine & Programming Language: Godot 4 with GDScript

- **Decision Rationale:** I selected Godot over Pygame because Godot provides native systems for scenes, tilemaps, animations, and state management. This allows me to focus my limited 4-week timeline on AI agent development and visual polish rather than building engine infrastructure from scratch.
- **Alternatives Dismissed:** Unity was discarded due to its steeper learning curve for a first engine project (C# and a high volume of simultaneous concepts). GameMaker was passed over because Godot offers similar capabilities while being entirely free and open-source.
- **Language Synergy:** GDScript was chosen because its syntax is heavily inspired by Python, allowing me to leverage my existing (though rusty) foundational knowledge.

- Artificial Intelligence: Tabular Q-learning for the Hunter

- **What will The Hunter do?** The Hunter is not gonna see the whole map, It will only know at what distance and direction the player is, the map it's divided in little squares (like a chess board).
- **The Actions (What he can do):** It only can move in 4 directions: *Up, down, left, right*.
- **The Reward (How it learns):**
 - positive reward for reducing the distance to the player
 - negative reward for increasing the distance
 - larger positive reward for catching the player
 - negative reward for invalid movement / hitting walls

- Visual Art Direction: High-Contrast Geometric Minimalism

- **Execution:** The game will be inspired from Frank-Miller-inspired noir comic-book aesthetic, utilizing a single 3-color palette (black and white + red accents), and it will be done using AI.
- **Technical Rationale:** This approach completely bypasses the need for advanced pixel art skills or complex frame-by-frame animation, keeping production time tightly constrained while matching the intended 1960s Noir comic-book aesthetic

4. Planning

The project will be developed over 4 weeks, following the phases of **Analysis, Design, Implementation, and Validation**, with game development and AI development progressing in parallel.

Week	Game Development	AI	Phase
1	Research which engine is most suitable and why, research the visual style, and finish the MoSCoW table to define the scope of the game.	Research Q-learning applied to grid-based pursuit and determine the initial approach.	Analysis.
2	Create the base map, player movement, guard patrols, and vision-based detection.	Design the architecture of The Hunter agent (Q-table, state representation, and reward function).	Design → Start of Implementation
3	Implement the visual direction (palette, silhouettes, contrast), and polish collisions and movement.	Implement and train The Hunter, iterating until measurable improvement can be observed.	Implementation
4	Final integration and general polishing.	Compare the trained Hunter with an untrained/random agent and visualize the learning curve.	Final Implementation + Validation